**Problem Statement 1: AI-Powered Omics Data Intelligence Platform**

**Title:**
Development of an AI-Based Integrated Omics Analytics System for Disease Risk Prediction

**Problem Statement:**
Large-scale omics datasets (genomics, transcriptomics, proteomics) generate high-dimensional biological data that are difficult to analyze using traditional computational methods. Participants are required to develop an AI-driven framework capable of managing, integrating, and analyzing multi-omics datasets to predict disease susceptibility and identify potential therapeutic targets.

**Expected Outcomes:**

- Feature selection from high-dimensional omics data
- Disease risk classification model
- Biomarker identification
- Model explainability using SHAP/LIME

**Problem Statement 2: AI-Based Toxicity and Adverse Reaction Prediction**

**Title:**
Predictive AI Model for Early Toxicity Screening of Biopharmaceutical Compounds

**Problem Statement:**
Drug toxicity remains a major cause of failure in clinical trials. Develop a machine learning or deep learning model that predicts toxicity and adverse reactions of biopharmaceuticals using molecular descriptors, biological response data, or gene-expression signatures.

**Expected Outcomes:**

- Toxicity classification (Safe / Toxic)
- Adverse reaction risk scoring
- Performance benchmarking with baseline ML models
- Interpretability of chemical/biological features

**Problem Statement 3: Mutation Impact & Disease Association Prediction**

**Title:**
AI-Based Prediction of Functional Impact of SNPs and Genetic Mutations

**Problem Statement:**
Single Nucleotide Polymorphisms (SNPs) and genetic mutations play a critical role in disease progression. Build an AI model to predict whether a given mutation is benign or disease-associated and assess its functional impact.

**Expected Outcomes:**

- Mutation classification model
- Functional impact scoring
- Integration with genomic databases
- Visualization of mutation-pathway association

**Problem Statement 4: AI for Antimicrobial Resistance (AMR) Prediction**

**Title:**
Machine Learning Framework for Predicting Antimicrobial Resistance Patterns

**Problem Statement:**
Antimicrobial resistance is a global health crisis. Develop an AI-based model that predicts resistance patterns of pathogens using genomic sequences, hospital data, or drug exposure history.

**Expected Outcomes:**

- Pathogen-drug resistance prediction
- Resistance trend analysis
- Risk stratification dashboard
- Model validation with real-world datasets

**Problem Statement 5: AI-Assisted Vaccine & Antibody Engineering**

**Title:**
Computational Design of High-Stability Antibody Candidates Using AI

**Problem Statement:**
Design an AI-based system capable of predicting protein stability and antigen-binding efficiency to assist in vaccine and antibody engineering.

**Expected Outcomes:**

- Protein stability prediction
- Antigen-antibody interaction modeling
- Mutational optimization suggestions
- Visualization of protein structure impact

**Problem Statement 6: Host–Pathogen Interaction Prediction**

**Title:**
Deep Learning Model for Predicting Host–Pathogen Molecular Interactions

**Problem Statement:**
Understanding host-pathogen interactions is critical for infectious disease control. Develop a predictive model using protein sequence data or interaction networks to identify potential host-pathogen binding interactions.

**Expected Outcomes:**

- Interaction prediction model
- Network visualization
- Identification of key interaction hubs
- Model explainability

**Problem Statement 7: AI-Based Biological Pathway Modeling**

**Title:**
AI Framework for Simulation and Analysis of Biological Pathways

**Problem Statement:**
Biological pathways are complex networks regulating cellular processes. Develop an AI-driven simulation or network analysis tool to predict pathway behavior under different perturbations.

**Expected Outcomes:**

- Network construction model
- Simulation of pathway perturbations
- Identification of regulatory nodes
- Visualization dashboard

**Problem Statement 8: LLM-Based Self-Learning Biotechnology Assistant**

**Title:**
AI-Powered Personalized Biotechnology Learning Platform

**Problem Statement:**
Develop a Large Language Model (LLM)-based interactive learning assistant that generates personalized self-learning modules for biotechnology students at school or university level.

**Expected Outcomes:**

- Adaptive learning content generation
- Quiz & assessment module
- Personalized learning path

- Feedback analytics dashboard

**Problem Statement 9:Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases in Coimbatore District**

**Problem Statement 9**

Water-borne diseases such as cholera, typhoid, dysentery, and hepatitis A remain a recurring public health concern in Coimbatore District, particularly during monsoon seasons and in areas with inadequate sanitation and unsafe drinking water. Delayed reporting from hospitals and primary health centers, lack of integration between water quality data and health surveillance systems, and absence of predictive analytics lead to reactive rather than preventive healthcare interventions.

**Expected Outcomes:**

- AI-based outbreak prediction model (7–14 days in advance)
- Risk heatmap for wards/villages
- Automated early warning alert system
- Interactive monitoring dashboard
- Scalable and deployable solution for real-world implementation

**Problem Statement 10: Generative Adversarial Network-Based Computational Design of Stable Antibody Candidates**

**Problem Statement:**

Design a GAN-driven AI platform capable of generating novel protein sequences with enhanced stability and antigen-binding affinity. The system should leverage adversarial learning to simulate mutational effects, optimize structural conformations, and accelerate vaccine and antibody engineering processes while reducing laboratory dependency.

**Expected Outcomes:**

• GAN-based protein sequence generation
• Stability and antigen-binding affinity prediction
• Mutational optimization recommendations
• Structural impact visualization dashboard
• Reduced experimental validation time

**Problem Statement 11:Blockchain-Based Supply Chain Transparency Platform with Generative AI Insights**

**Problem Statement:**

Develop a blockchain-enabled agricultural supply chain tracking system integrated with Generative AI models to generate predictive demand analytics, anomaly detection reports, and risk forecasting. The system should ensure traceability while leveraging generative intelligence for proactive logistics optimization.

**Expected Outcomes:**

• Immutable blockchain traceability
• AI-generated demand forecasting reports
• Fraud and anomaly detection analytics
• Real-time supply chain dashboards
• Improved producer-to-consumer transparency

**Problem Statement 12;GAN-Driven Integrated Multi-Omics Analytics System for Disease Risk Prediction**

**Problem Statement:**

Develop a GAN-based AI framework capable of integrating genomics, transcriptomics, and proteomics datasets to predict disease susceptibility. The system should generate synthetic omics samples for rare conditions, enhance feature extraction through adversarial learning, and improve biomarker discovery accuracy.

**Expected Outcomes:**

• Synthetic multi-omics data augmentation
• Adversarial feature selection
• Disease risk classification model
• Biomarker discovery through latent representation analysis
• Model explainability using SHAP/LIME

**Problem Statement 13:Generative AI Conversational Interface for ARGO Ocean Data Discovery (FloatChat)**

**Problem Statement:**

Design a Generative AI-powered conversational system that enables natural language querying, automated summarization, and visualization of oceanographic datasets. The assistant should transform user prompts into analytical workflows and generate interpretable marine data insights.

**Expected Outcomes:**

• Natural language ocean data querying
• AI-generated summaries of ocean trends
• Automated visualization dashboards
• Predictive marine climate analytics
• Enhanced accessibility for researchers

**Problem Statement 14:Generative Adversarial Learning Framework for Predicting Antimicrobial Resistance Patterns**

**Problem Statement:**

Develop a GAN-based system that generates synthetic pathogen genomic sequences to enhance antimicrobial resistance prediction. The adversarial model should detect emerging resistance mutations and improve generalization across unseen drug-pathogen combinations.

**Expected Outcomes:**

• Pathogen-drug resistance prediction model
• Synthetic AMR data generation
• Resistance trend forecasting
• Risk stratification dashboard
• Validation using real-world datasets

**Problem Statement** 15:Generative AI-Based API Integration of NAMASTE & ICD-11 TM2 into EHR-Compliant EMR Systems

**Problem Statement:**

Create secure API middleware leveraging Generative AI to integrate standardized clinical coding frameworks into EHR-compliant EMR systems. The system should auto-generate diagnostic code suggestions and enhance interoperability between traditional and modern medical records.

**Expected Outcomes:**

• Seamless API integration
• AI-generated clinical coding recommendations
• Enhanced healthcare interoperability
• Compliance with Indian EHR standards
• Improved documentation efficiency

**Problem Statement** 16;:Autonomous Precision Planting and Weeding Machine Using AI and Generative Simulation

**Problem Statement:**

Develop an autonomous agricultural robot equipped with AI-based vision systems and Generative AI simulation models to optimize crop planting density and precision weed removal with minimal ecological disruption.

**Expected Outcomes:**

• AI-based crop and weed detection
• Generative simulation for optimal planting strategies
• Autonomous navigation system
• Reduced chemical usage
• Increased agricultural productivity

**Problem Statement 17:Generative AI-Driven Next-Generation Firewall with Zero Trust Implementation**

**Problem Statement:**

Design an AI-powered cybersecurity firewall integrated with Generative AI-based threat modeling to dynamically generate adaptive security rules and detect evolving cyber threats within Zero Trust architectures.

**Expected Outcomes:**

• AI-based anomaly detection
• Generative threat intelligence models
• Dynamic security rule creation
• Automated Zero Trust enforcement
• Improved enterprise security resilience

**Problem Statement 18: GAN-Based Toxicity Screening Framework for Biopharmaceutical Compounds**

**Problem Statement:**

Develop a GAN-driven toxicity prediction system capable of generating synthetic molecular descriptors and biological response patterns to enhance early toxicity detection and adverse reaction risk assessment.

**Expected Outcomes:**

• Toxicity classification (Safe/Toxic)
• Synthetic molecular data augmentation
• Adverse reaction risk scoring
• Performance benchmarking
• Feature interpretability mechanisms

**Problem Statement 19 :GAN-Based Early Warning System for Water-Borne Diseases in Coimbatore District**

**Problem Statement:**

Develop a GAN-based smart health surveillance system that integrates environmental monitoring, water quality data, and clinical case reports. The adversarial model should simulate outbreak scenarios and improve early detection accuracy for water-borne diseases.

**Expected Outcomes:**

• Synthetic outbreak scenario simulation
• Risk prediction model for early warning
• Environmental-health data augmentation
• Community alert system
• Interactive decision-support dashboard

**Problem Statement 20:GAN-Based Mutation Impact and SNP Disease Association Prediction**

**Problem Statement:**

Create a GAN-based AI model capable of generating mutated genomic sequences and learning adversarial representations to classify mutations as benign or disease-associated while estimating functional impact scores.

**Expected Outcomes:**

• Mutation classification model
• Functional impact scoring
• Synthetic mutation data generation
• Integration with genomic databases
• Pathway association visualization

**Problem Statement 21:Generative AI-Powered FRA Atlas and WebGIS Decision Support System**

**Problem Statement:**

Develop a WebGIS-based decision support system integrated with Generative AI to analyze forest rights spatial data, generate compliance summaries, and predict land-use conflicts for improved governance transparency.

**Expected Outcomes:**

• AI-generated compliance reports
• Interactive WebGIS visualization
• Predictive conflict detection
• Policy decision support tools
• Transparent governance analytics

**Problem Statement 22 :AI-Based Dynamic Contact Resistance Measurement (DCRM) Analysis with Generative Diagnostics**

**Problem Statement:**

Design a Generative AI-integrated diagnostic framework to analyze DCRM waveforms in EHV circuit breakers, detect latent fault signatures, and generate predictive maintenance recommendations.

**Expected Outcomes:**

• Automated waveform analysis
• AI-generated fault diagnostics
• Predictive maintenance alerts
• Grid reliability enhancement
• Reduced maintenance cost

**Problem Statement 23: GAN-Based Simulation Framework for Biological Pathway Modeling**

**Problem Statement:**

Develop a GAN-driven simulation system that generates synthetic pathway perturbations and learns dynamic regulatory behavior of biological networks under stress or mutation conditions.

**Expected Outcomes:**

• Synthetic pathway state generation
• Network perturbation simulation
• Identification of key regulatory nodes
• Visualization dashboard
• Improved pathway prediction accuracy.